FW 0.8.6-0_2026-10-30T03:50:10Z | test 2026-10-30 04:21:46 to 05:07:30 UTC | 400 sps | logger 99977

Folder: 2026-10-30_FW0860_deck_PPS_400sps
Pulses analyzed: ADC1 2744, ADC2 2744, ADC3 2744, ADC4 2744

Timing (negative offset = recorded edge later than reference; RMS resid = raw/after per-pulse shift)

ADC	chan	n	offset med (std) ms	theory ms	obs-theory	best-fit shift med (std) ms	RMS resid %
ADC1	HDF	2744	-0.302 (0.018)	-0.000	-0.302	-0.221 (0.031)	1.04/ 0.62
ADC2	EHZ	2744	-7.048 (0.712)	-5.994	-1.054	-1.078 (0.712)	2.61/ 0.30
ADC3	EHN	2744	-6.866 (0.713)	-5.994	-0.872	-0.889 (0.712)	2.31/ 0.30
ADC4	EHE	2744	-6.715 (0.709)	-5.994	-0.721	-0.730 (0.708)	2.08/ 0.32

ADC clocks relative to ADC1 (from the sawtooth of the ADC2-4 offsets)

ADC	chan	clock - ADC1, ppm: median	5-95% over run	ADC clock vs TCXO, ppm
ADC2	EHZ	-289 (alias limit ±1250)	-316 ... -271	+14
ADC3	EHN	+534 (alias limit ±1250)	+514 ... +553	+837
ADC4	EHE	+692 (alias limit ±1250)	+632 ... +714	+995

ADC1 clock vs TCXO (log adc_ppm median): +303 ppm. ADC1 offset trend: -39.9 us/hour.

How to read this report

- Theory = the signal that should have been recorded: ideal 1PPS step through the analog filter (ADC2-4 only) and the ADC sinc3 filter, sampled on the UTC grid, then passed through the same analysis as the data (levels, padding, aliasing).
- sinc3 is zero-phase: firmware time-tags each sample at the center of the sinc3 window.
- ADC2-4 analog model: ladder, corner 48.25 Hz per stage, stage-2 impedance 10 x stage 1.
- Edge plots: time axis is ((t + 0.5) mod 1) - 0.5 seconds from the UTC second. Spectra: DFT of each pulse relative to an ideal step (periodic square-wave padding), so only odd harmonics of 1 Hz exist; negative frequencies mirror positive ones; phase is unwrapped outward from low frequency. Envelope = mean ± 2 std over ALL pulses (<=150 individual drawn).
- Offset = phase slope through the origin below 40 Hz. Best-fit shift = time shift of the theoretical edge that minimizes squared error against each observed edge in the time domain.
- ADC2-4 are read at ADC1's DRDY while converting on their own clocks, so each sample is 0..T older than ADC1's (T = sample period): expected mean extra lag T/2 and offset scatter T/sqrt(12) = 0.722 ms (observed std: ADC2 0.712, ADC3 0.713, ADC4 0.709 ms).
- Clock beat: the offset sawtooth advances (clock difference x 1 s) per pulse, wrapped to +/- T/2. Sign: a faster ADC clock gives older data, so clock difference (ppm) = -(offset increment, ms/s) x 1000. Valid within the alias limit.

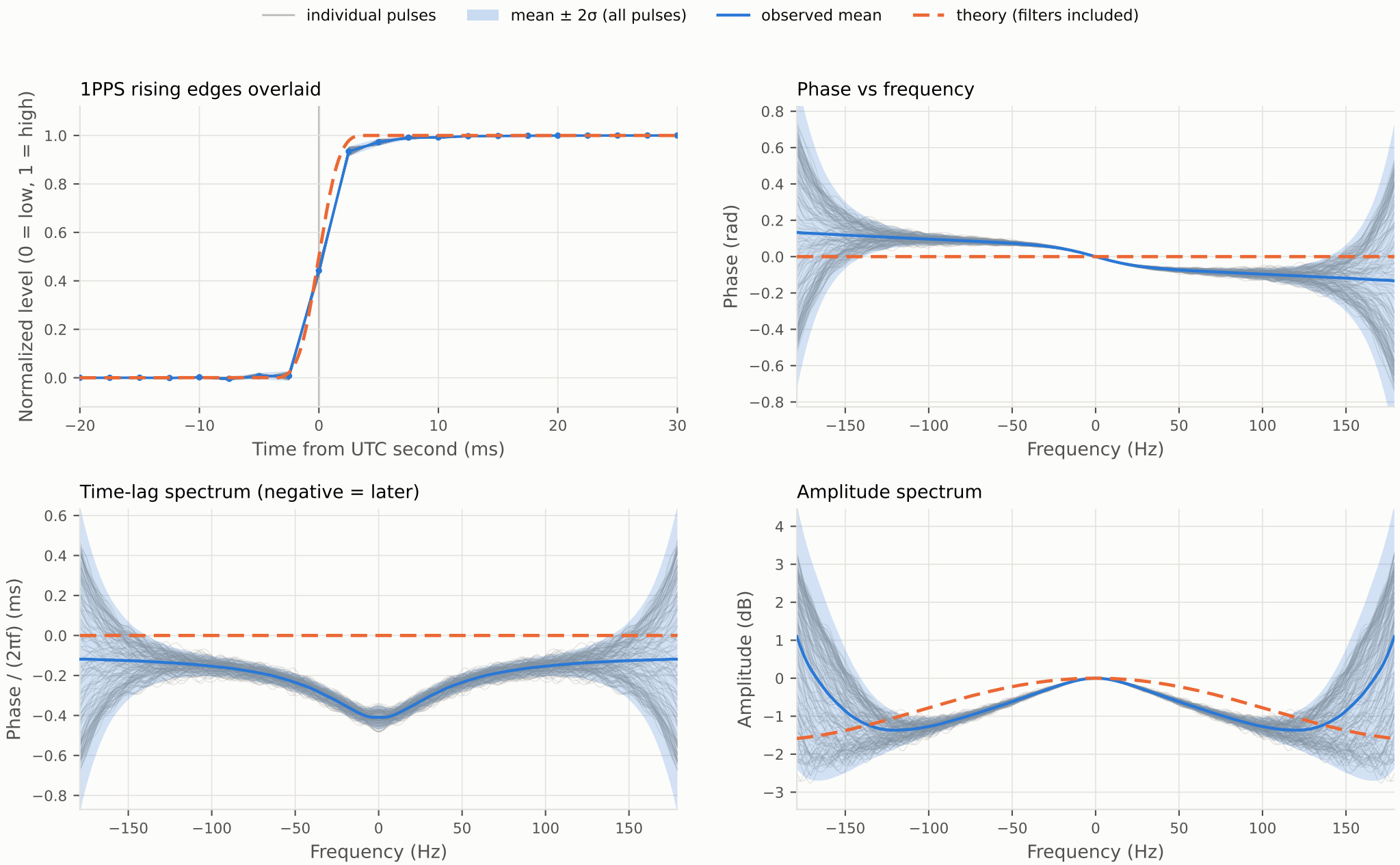
ADC1 (HDF): no analog filter + sinc3

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Time offset (negative = later), phase slope 0-40 Hz: observed median -0.302 ms (std 0.018) theory -0.000 ms observed - theory -0.302 ms

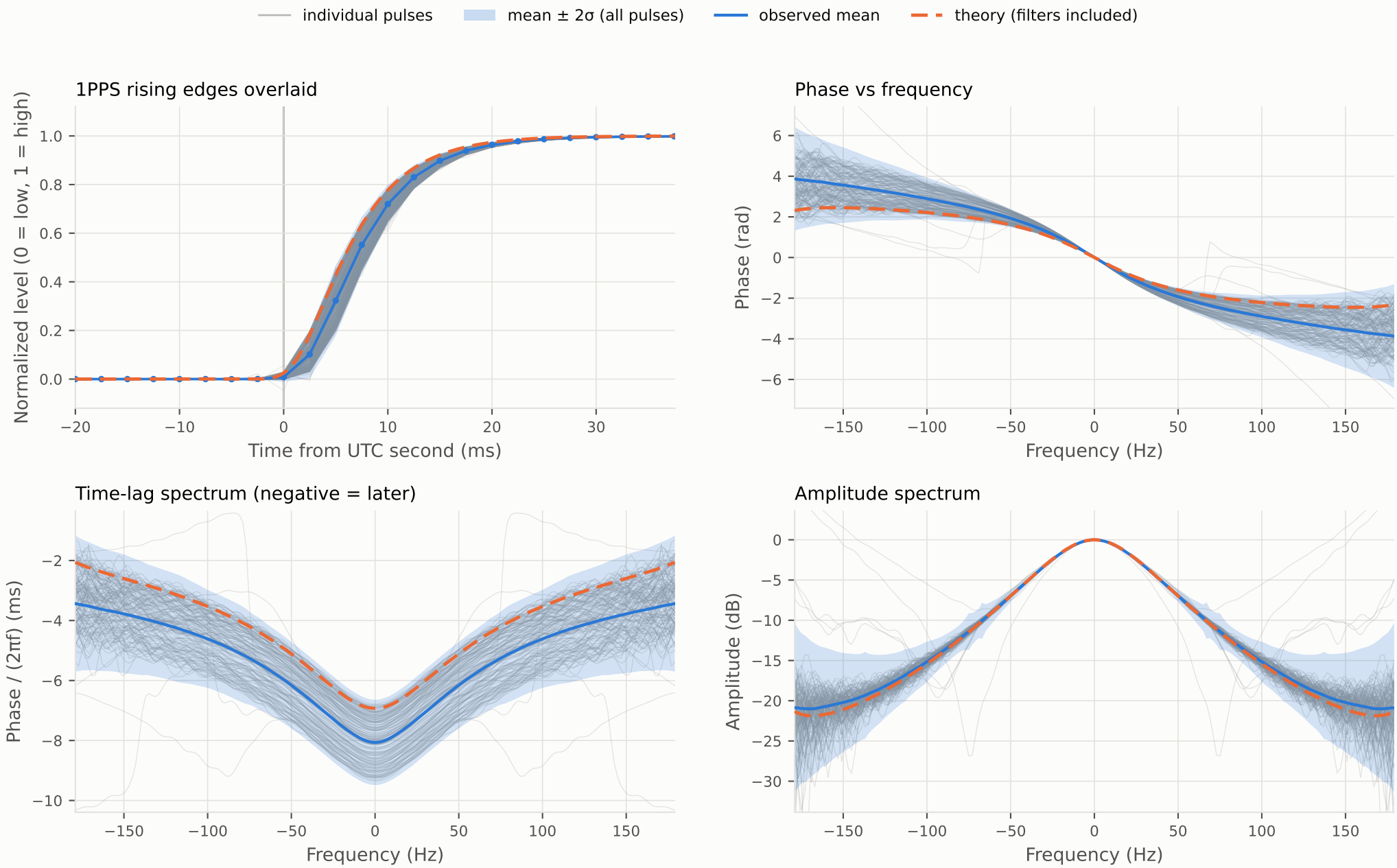
Per-pulse best-fit shift of theoretical edge: median -0.221 mean -0.221 std 0.031 2.5-97.5%: [-0.279, -0.170] ms

Edge residual RMS (% of step): 1.04 raw, 0.62 after per-pulse shift. Mean spectrum vs theory, RMS: phase 0.093 rad, amplitude 0.73 dB



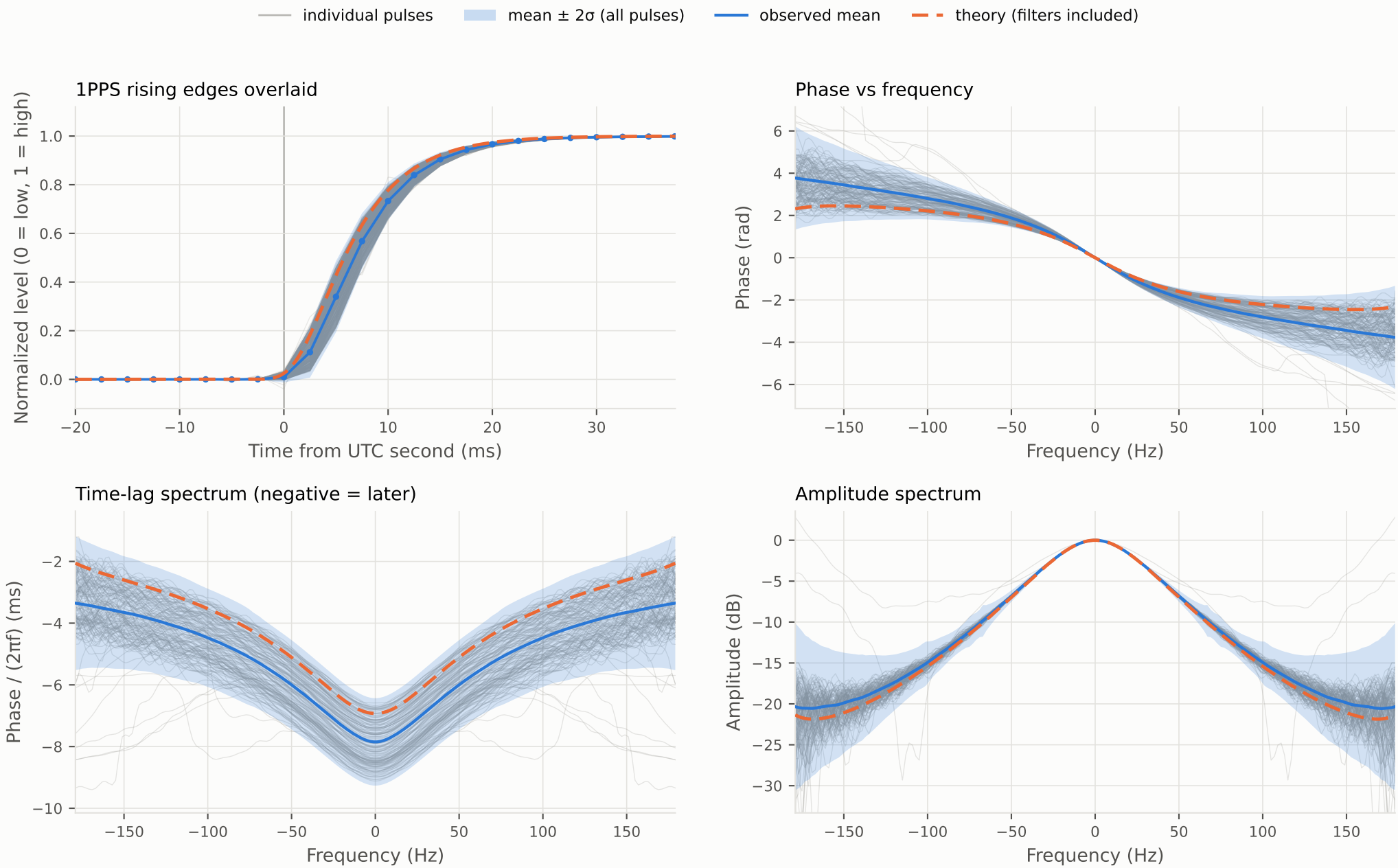
ADC2 (EHZ): 2nd-order passive RC ladder (Z ratio 10) 48.25 Hz/stage + sinc3

FW 0.8.6-0_2026-10-30T03:50:10Z | test 2026-10-30 04:21:46 to 05:07:30 UTC | 400 sps | logger 99977
Time offset (negative = later), phase slope 0-40 Hz: observed median -7.048 ms (std 0.712) theory -5.994 ms observed - theory -1.054 ms
Per-pulse best-fit shift of theoretical edge: median -1.078 mean -1.080 std 0.712 2.5-97.5%: [-2.255, +0.097] ms
Edge residual RMS (% of step): 2.61 raw, 0.30 after per-pulse shift. Mean spectrum vs theory, RMS: phase 0.769 rad, amplitude 0.46 dB



ADC3 (EHN): 2nd-order passive RC ladder (Z ratio 10) 48.25 Hz/stage + sinc3

FW 0.8.6-0_2026-10-30T03:50:10Z | test 2026-10-30 04:21:46 to 05:07:30 UTC | 400 sps | logger 99977
Time offset (negative = later), phase slope 0-40 Hz: observed median -6.866 ms (std 0.713) theory -5.994 ms observed - theory -0.872 ms
Per-pulse best-fit shift of theoretical edge: median -0.889 mean -0.889 std 0.712 2.5-97.5%: [-2.070, +0.287] ms
Edge residual RMS (% of step): 2.31 raw, 0.30 after per-pulse shift. Mean spectrum vs theory, RMS: phase 0.689 rad, amplitude 0.70 dB



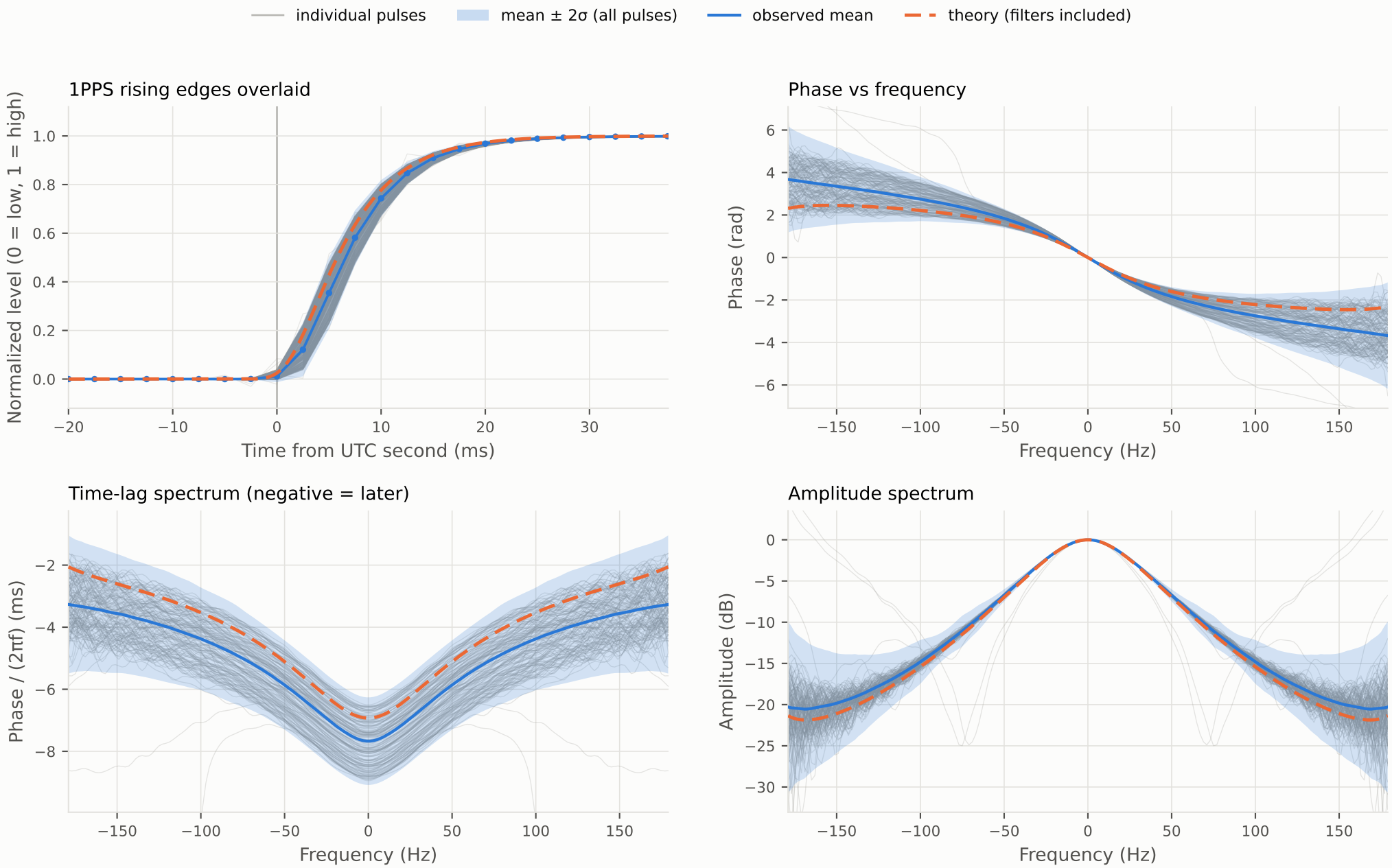
ADC4 (EHE): 2nd-order passive RC ladder (Z ratio 10) 48.25 Hz/stage + sinc3

FW 0.8.6-0_2026-10-30T03:50:10Z | test 2026-10-30 04:21:46 to 05:07:30 UTC | 400 sps | logger 99977

Time offset (negative = later), phase slope 0-40 Hz: observed median -6.715 ms (std 0.709) theory -5.994 ms observed - theory -0.721 ms

Per-pulse best-fit shift of theoretical edge: median -0.730 mean -0.730 std 0.708 2.5-97.5%: [-1.898, +0.430] ms

Edge residual RMS (% of step): 2.08 raw, 0.32 after per-pulse shift. Mean spectrum vs theory, RMS: phase 0.627 rad, amplitude 0.76 dB



Time offset and ADC clock beat vs time

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Offsets from the phase slope below 40 Hz (negative = recorded edge later); dashed gray = theory.

